

CardamomOT — analysis report

/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Kameneva

Generated on 2026-10-01 13:58

Run

Split	full
Stimulus penalisation	0.2
Prior penalisation	1.0
Genes	97
Stimuli	1
Observed cells	1800
Timepoints	0, 8, 32, 48, 64, 80
Cell types	precursors, intermediate, chromaffin, sympathoblasts
Network shown	inter_simul.npy, network #0 / 1
UMAP preprocessing	normalise=False, log1p=True

Perturbations (Data/KO_OV_simulate.txt)

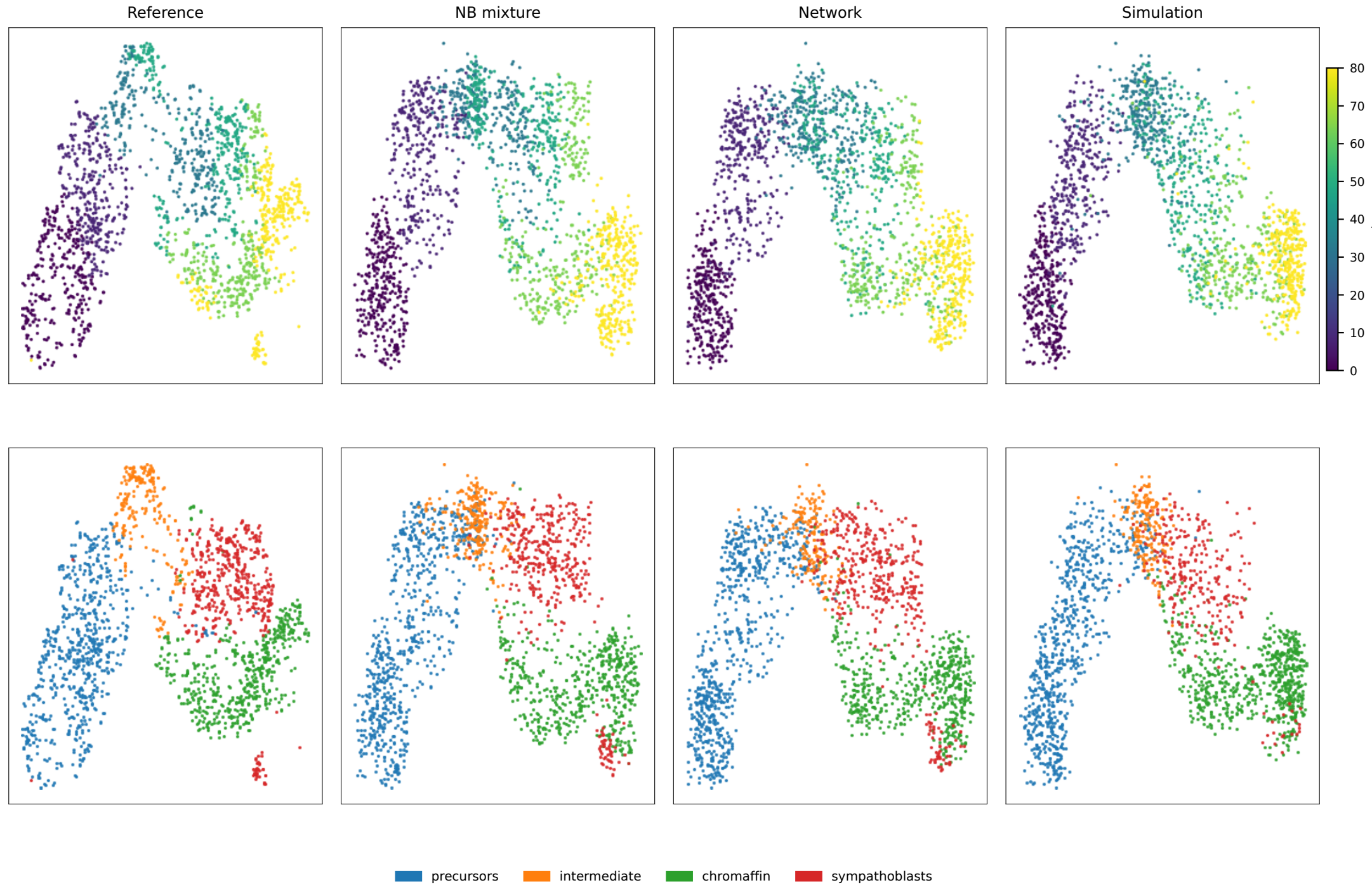
- ✓ KO CHGA (50%)
- ✓ KO CHGA
- ✓ KO CHGA · OV STMN2
- ✓ OV STMN2

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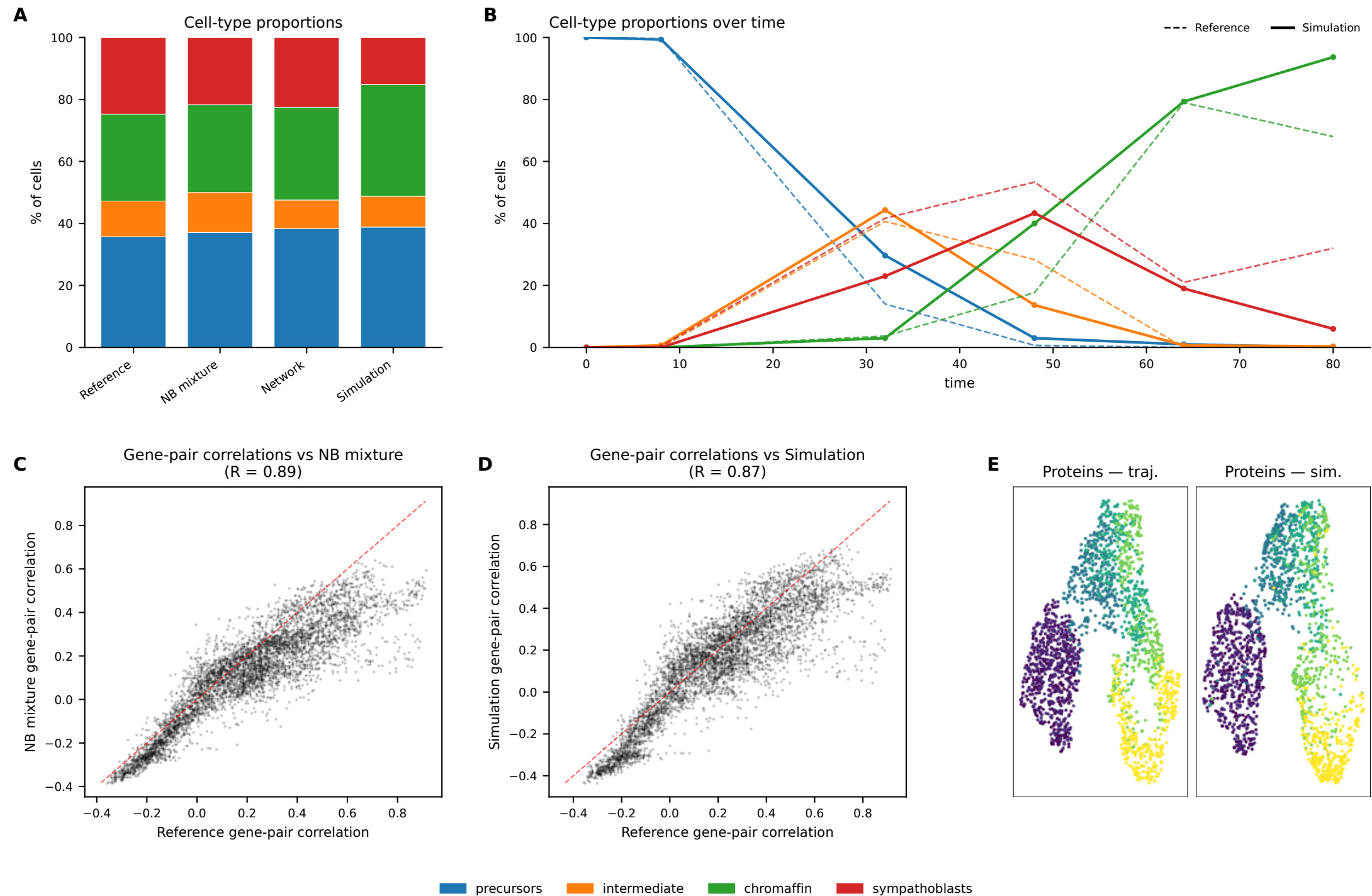
1. Generative model — UMAPs of data, NB mixture, network modes and simulation; cell-type proportions; gene-pair correlations; proteins
2. Gene regulatory network — regulatory power (violin plots) and top-10 regulators
3. In-silico perturbations — overview across KO/OV, then one page per perturbation
4. Proliferation — prior vs learned net rates, population growth, proteins driving growth
5. Learned dynamics — mRNA and protein velocity fields (mechanistic and along trajectories)

1. Generative model — trajectories and simulation

Joint UMAP of the observed data (Reference), NB mixture, network-driven modes and full simulation.



1. Generative model — quantitative checks

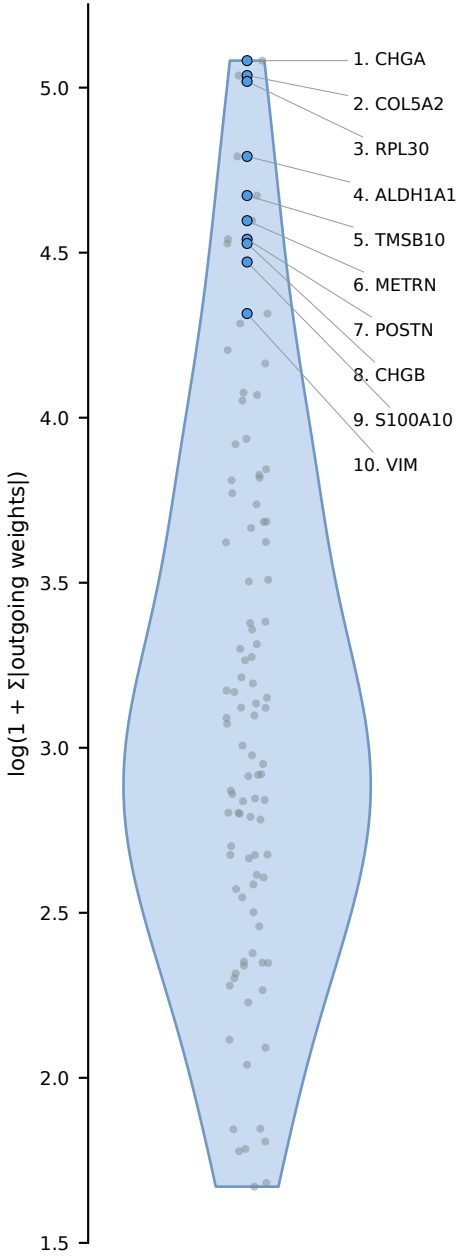


2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 10 genes labelled; stimulus not shown (stimulus = 0.2 < 0.5).

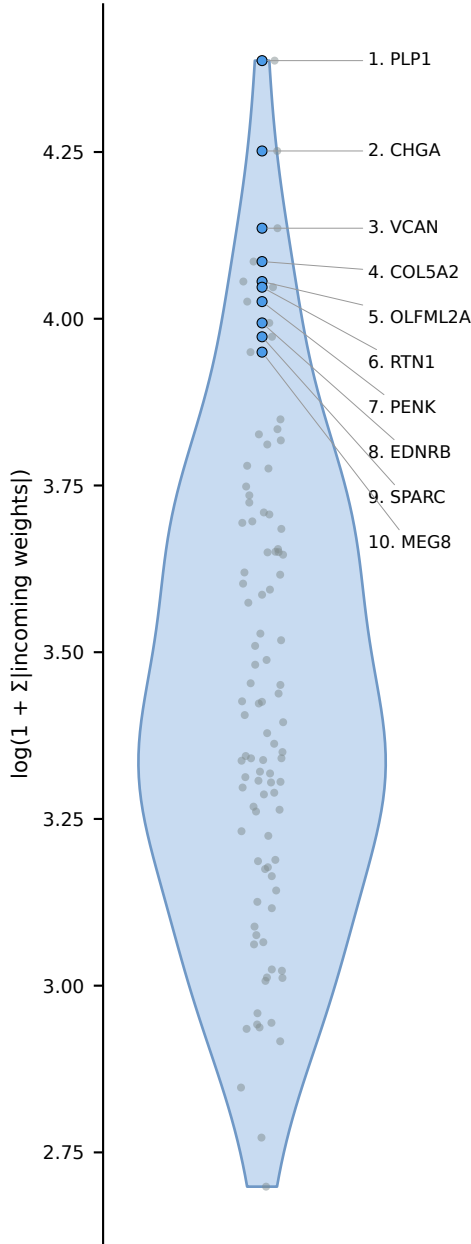
A

Outgoing regulation (regulators)



B

Incoming regulation (targets)



C

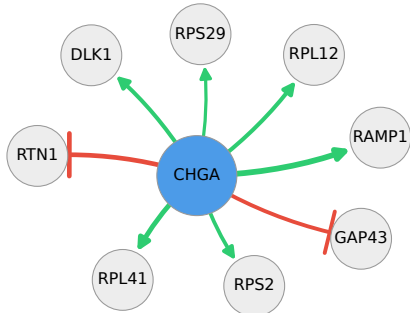
Top regulators

#	Regulator	$\sum w $	# act.	# inh.
1	CHGA	160.07	47	17
2	COL5A2	152.95	1	53
3	RPL30	150.24	70	9
4	ALDH1A1	119.48	7	62
5	TMSB10	106.02	59	14
6	METRN	98.18	24	13
7	POSTN	92.75	6	55
8	CHGB	91.55	29	17
9	S100A10	86.51	22	22
10	VIM	73.85	17	27

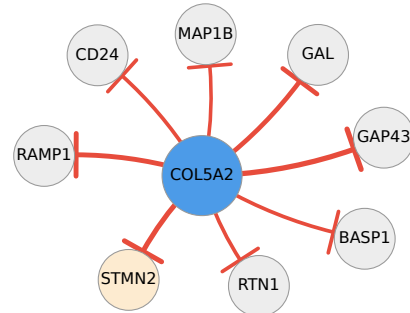
2. Gene regulatory network — top regulators and their main targets

Top 8 targets per regulator; edge width $\propto$ |weight| (global scale). Perturbed genes (KO/OV) are highlighted in beige.

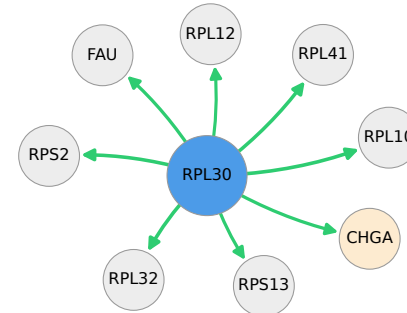
#1 CHGA



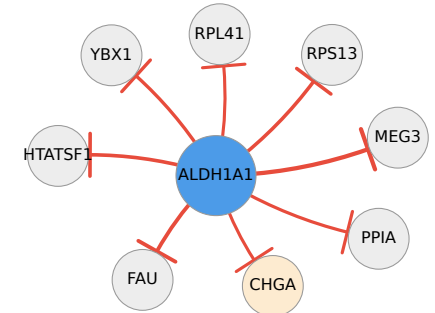
#2 COL5A2



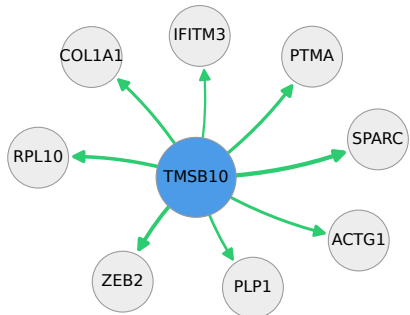
#3 RPL30



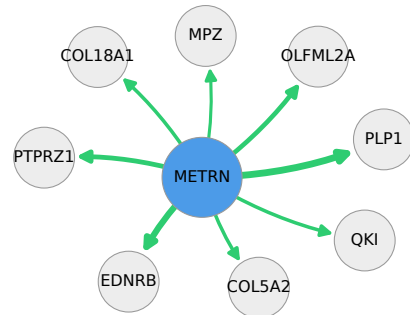
#4 ALDH1A1



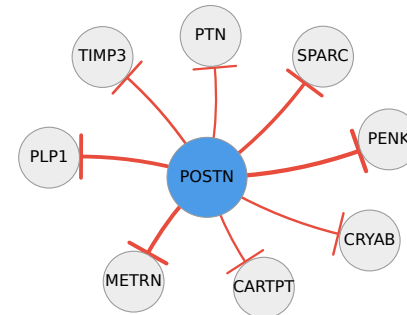
#5 TMSB10



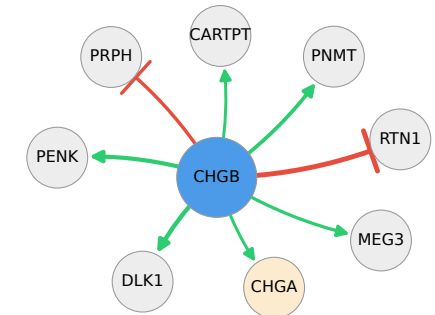
#6 METRN



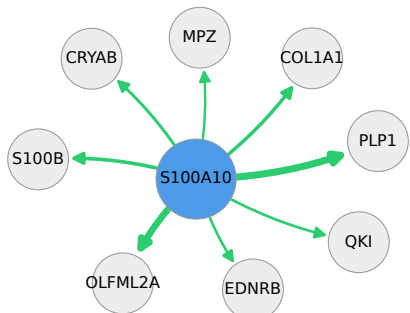
#7 POSTN



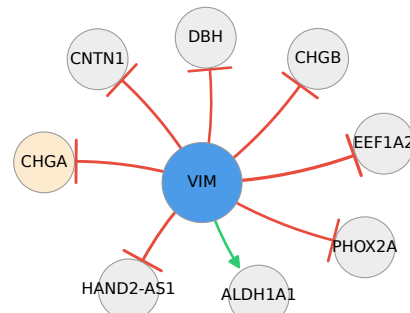
#8 CHGB



#9 S100A10



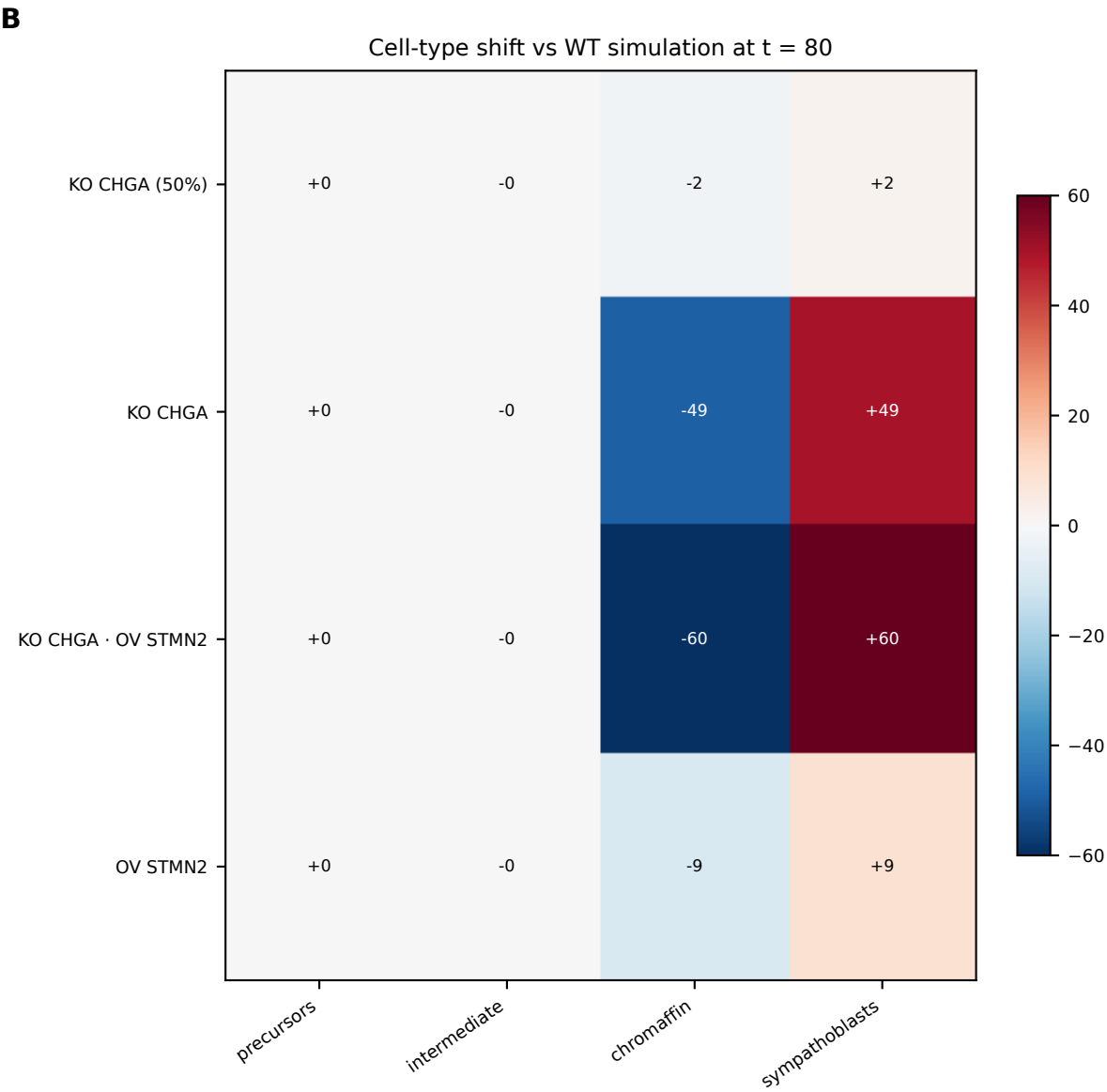
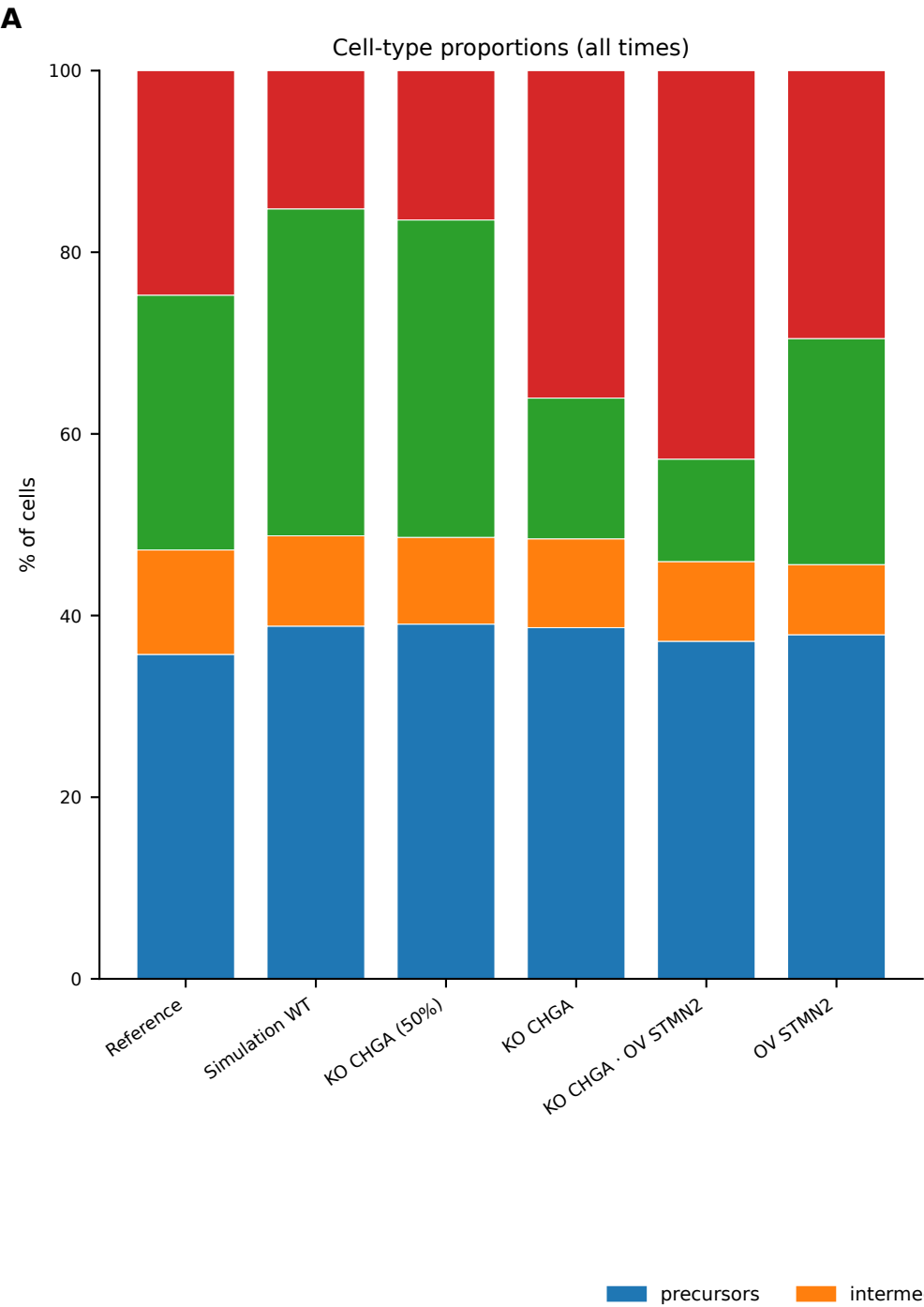
#10 VIM



— Activation — Inhibition — Top regulator

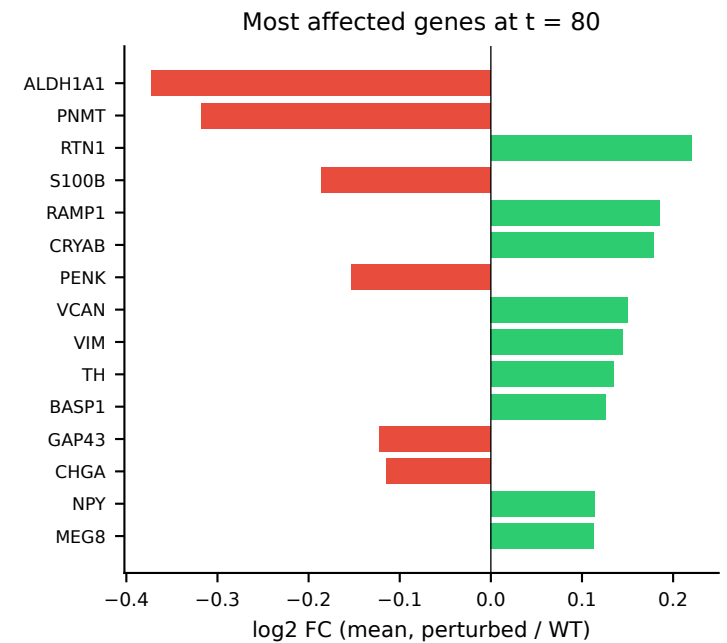
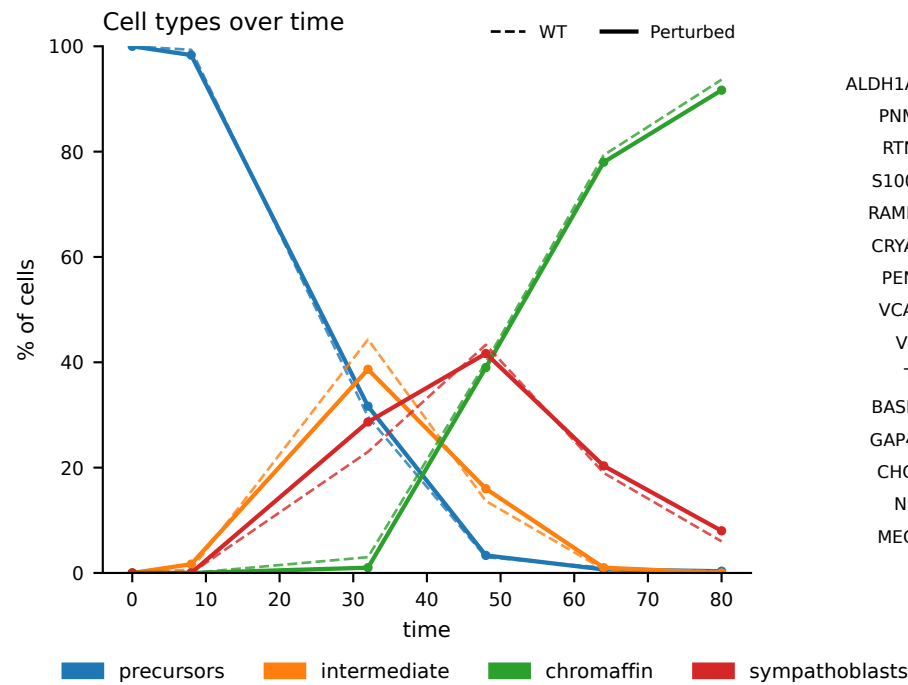
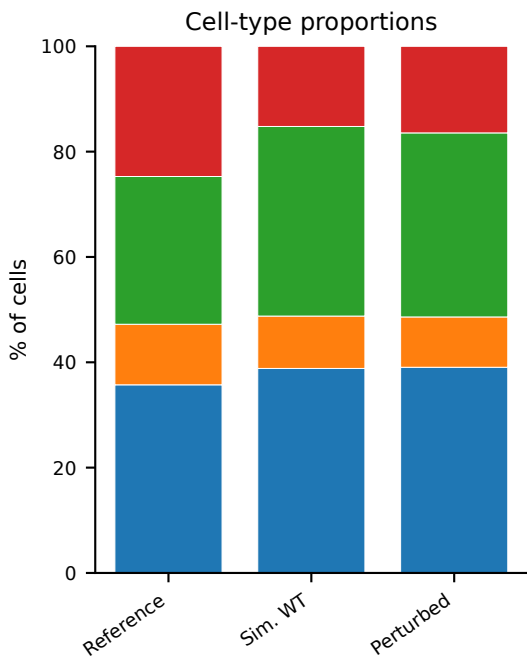
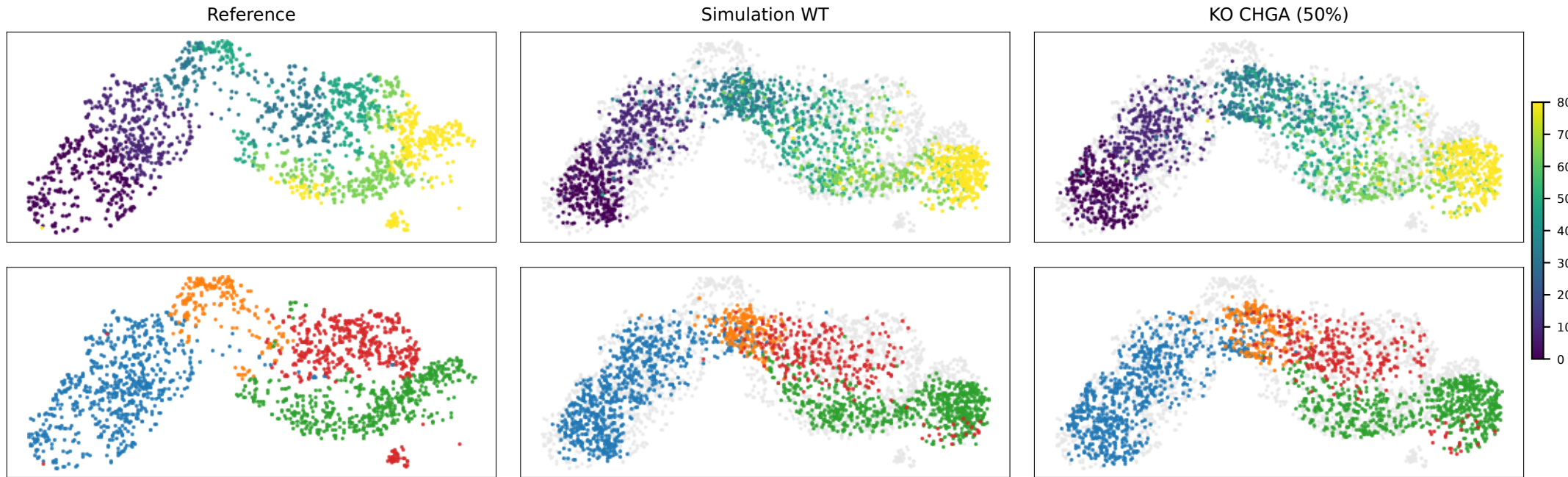
3. In-silico perturbations — overview

Cell types predicted by a random forest trained on the observed data.



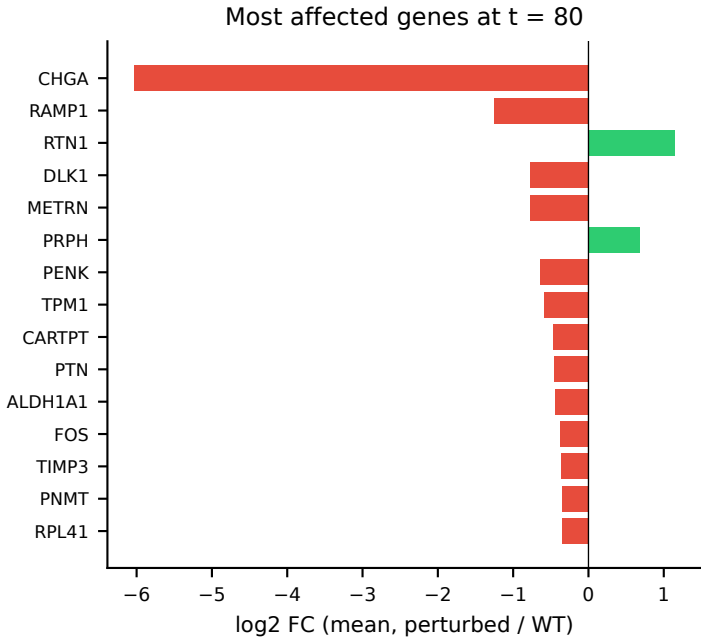
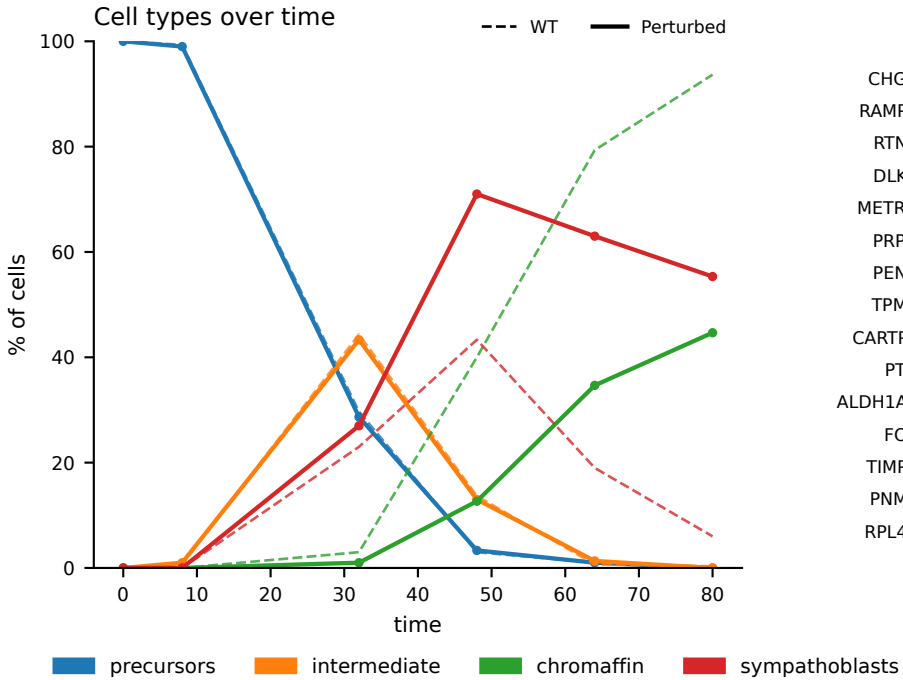
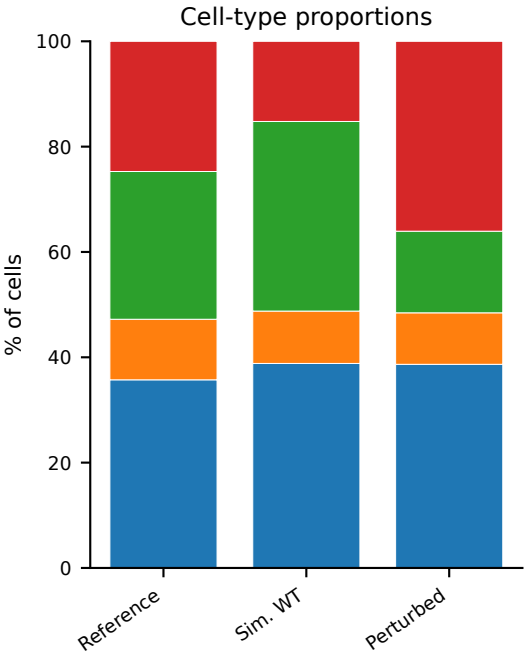
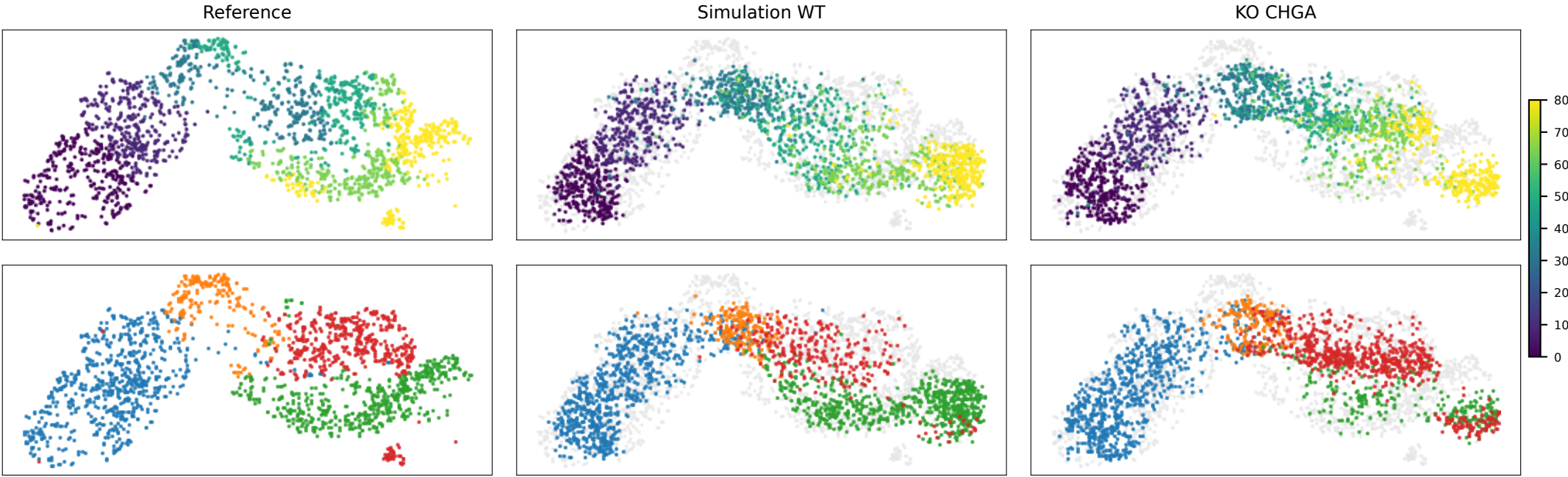
3. Perturbation — KO CHGA (50%)

KO_CHGApct50_OV_none. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



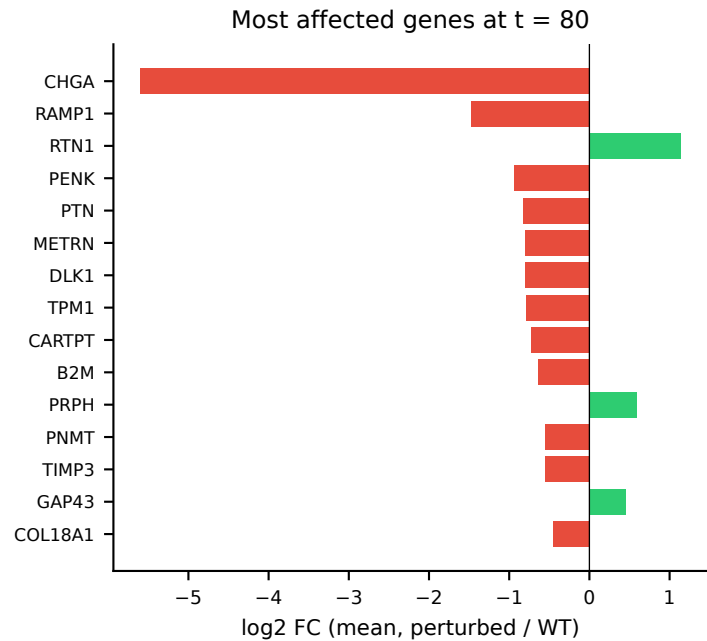
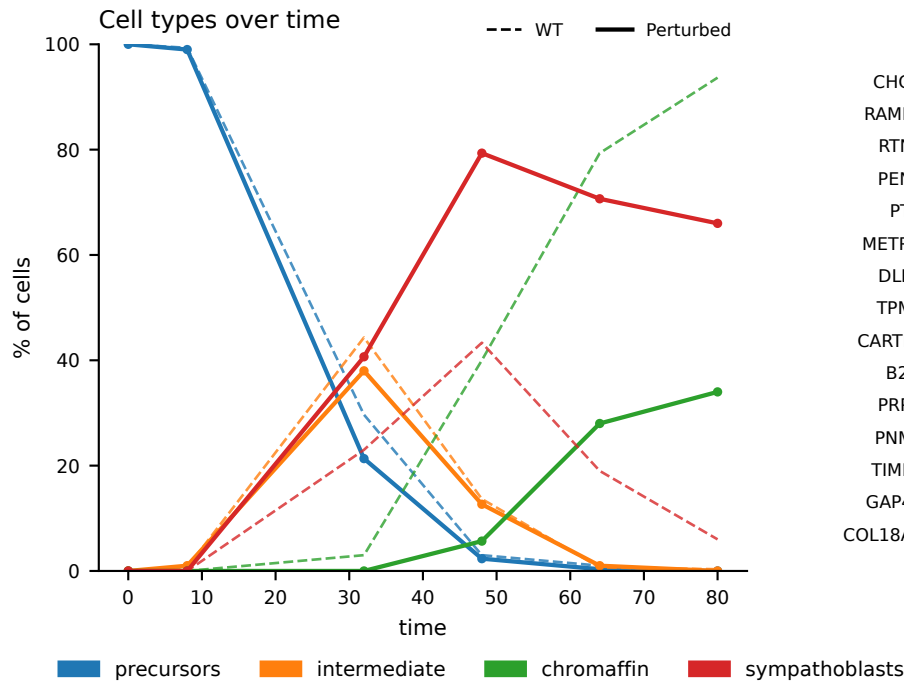
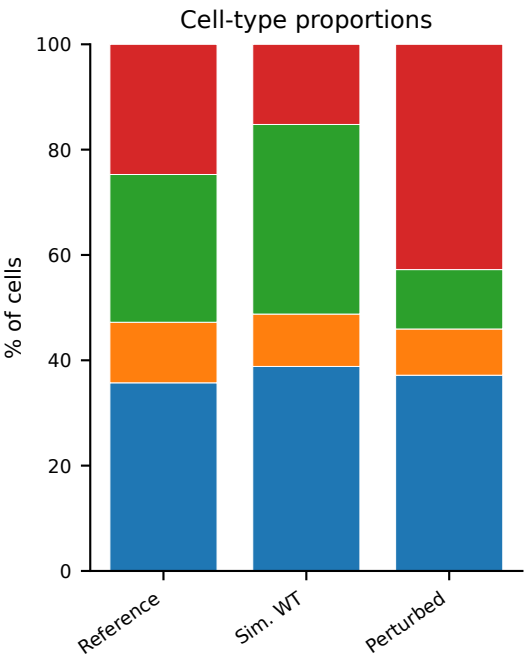
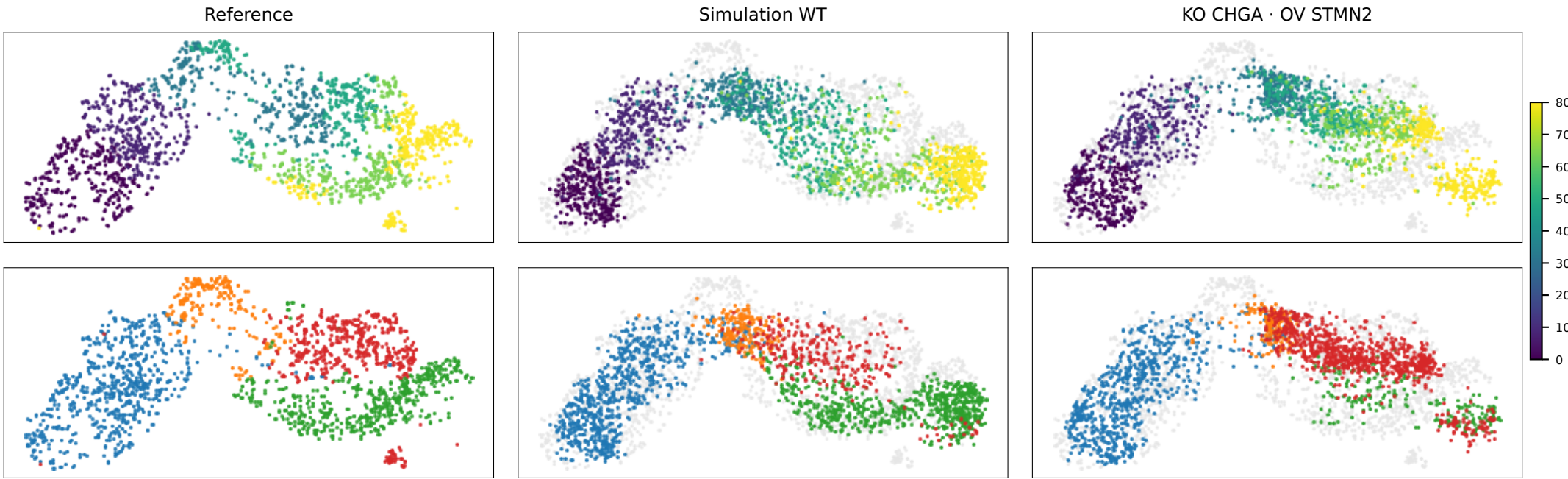
3. Perturbation — KO CHGA

KO_CHGA_OV_none. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



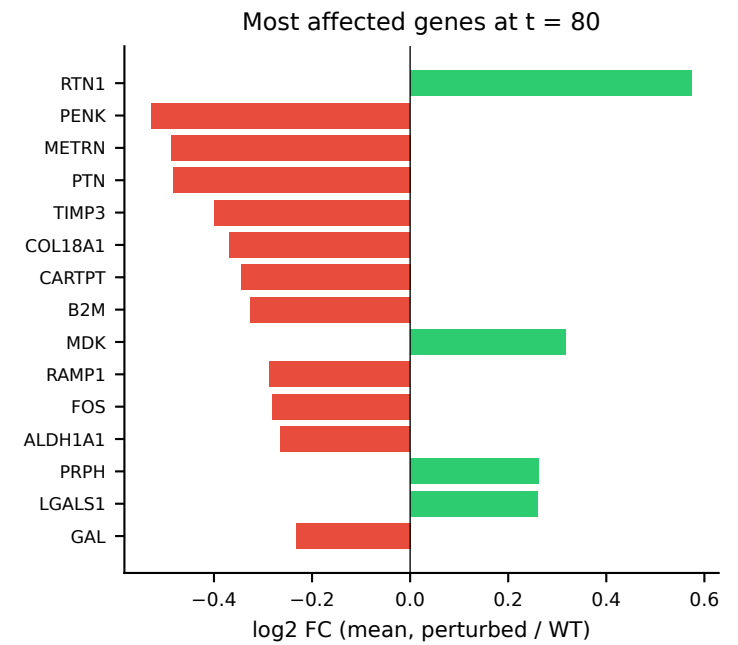
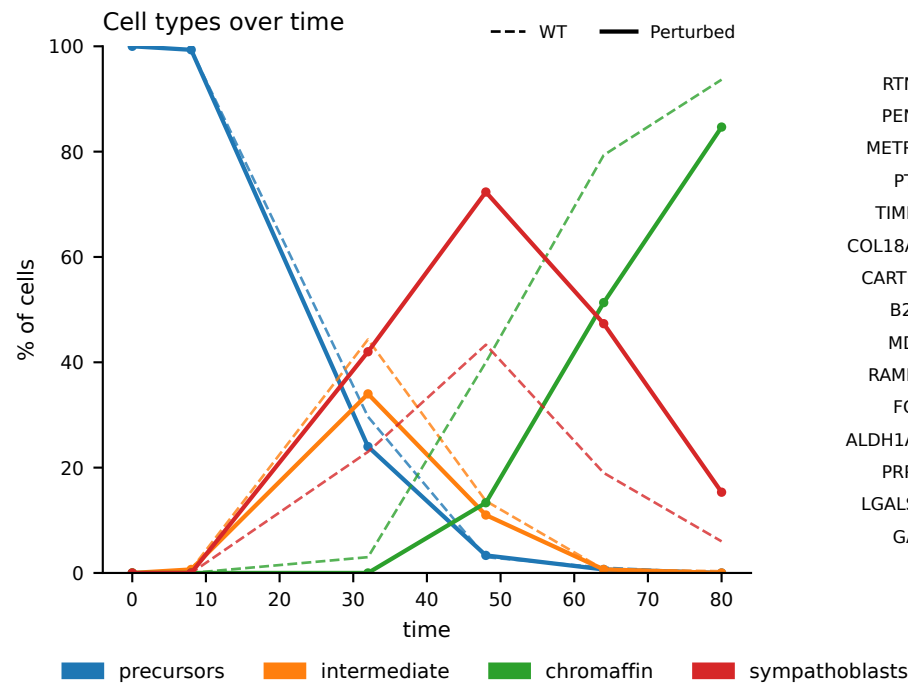
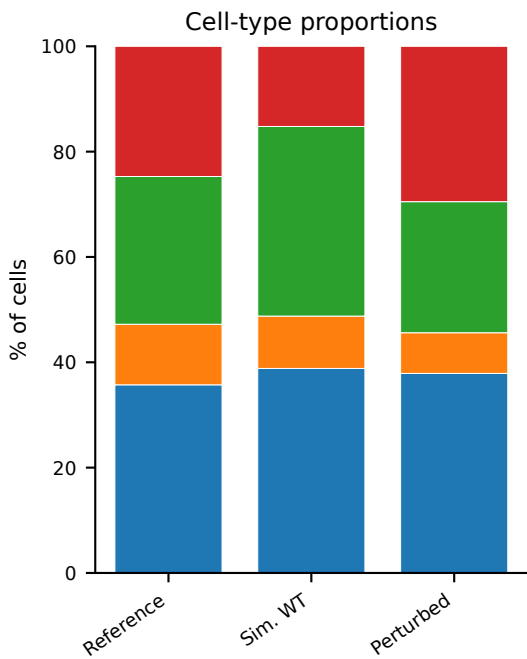
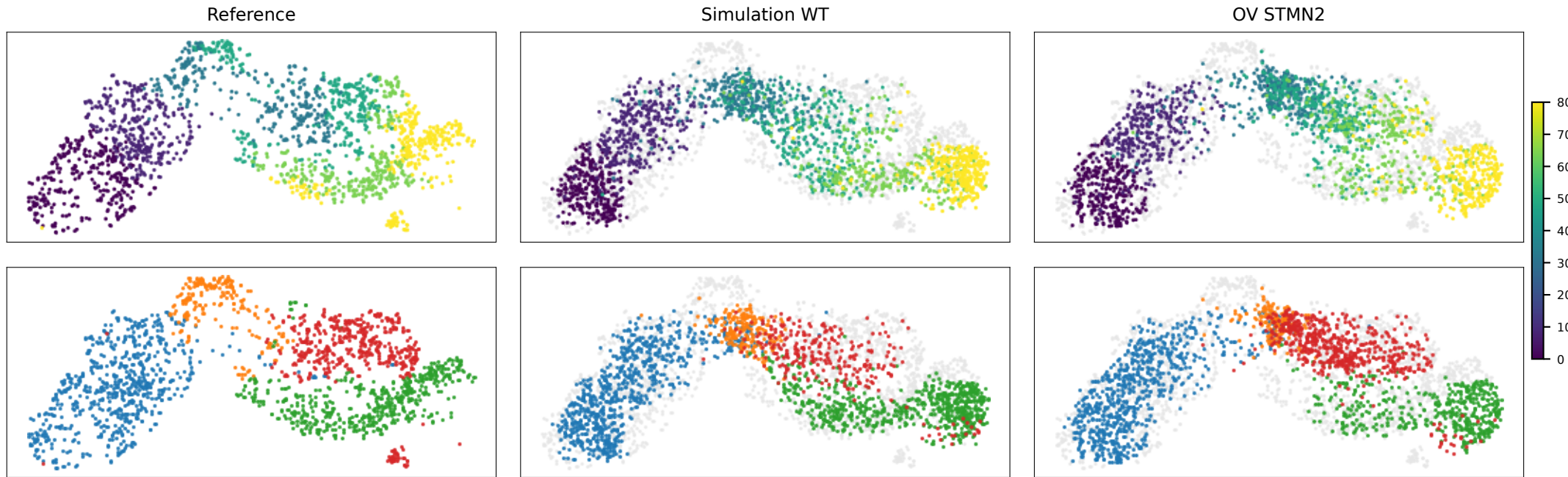
3. Perturbation — KO CHGA · OV STMN2

KO_CHGA_OV_STMN2. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



3. Perturbation — OV STMN2

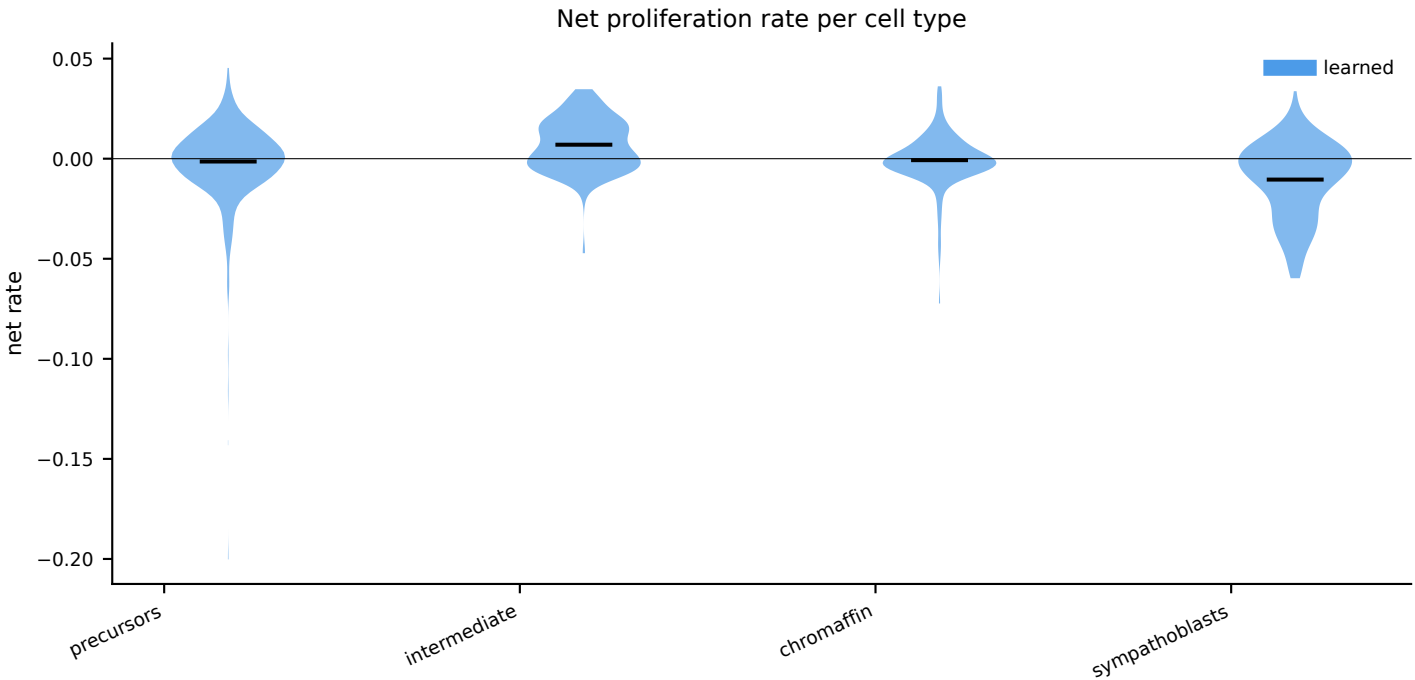
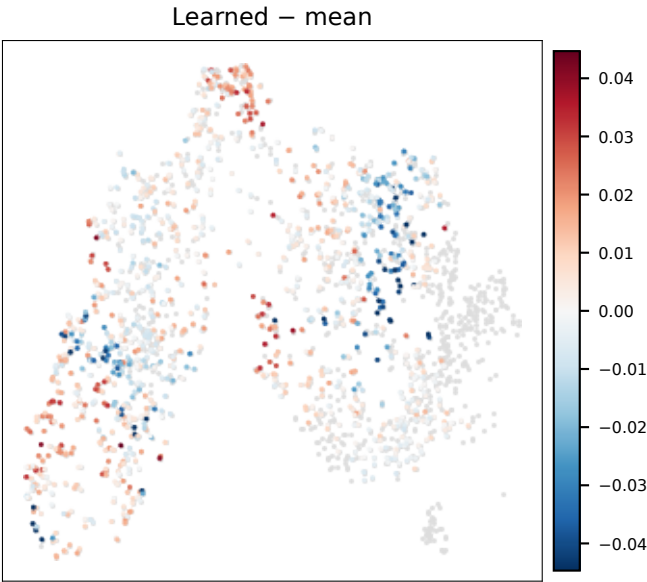
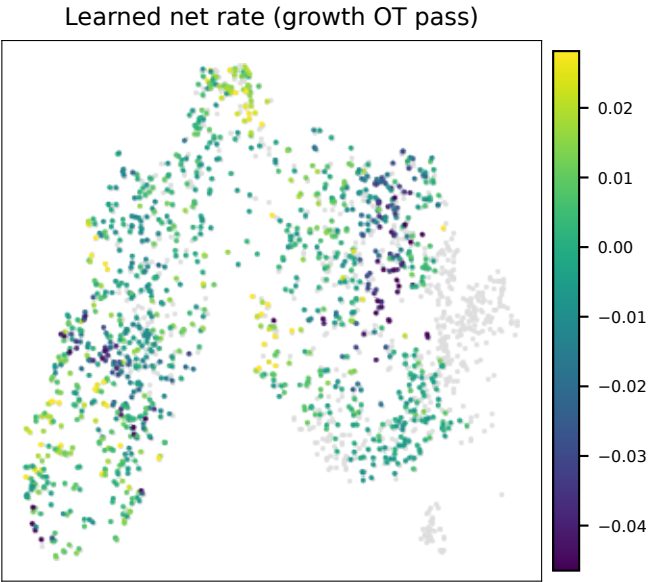
KO_none_OV_STMN2. UMAP: perturbed simulation projected on the WT embedding (grey = observed data).



4. Proliferation — prior vs learned net rate (birth – death)

Learned = growth OT pass, mean over the trajectory states behind each observed cell (grey = cell not reached). Rates per time unit of the data.

No prior provided:
uniform net rate
(only relative rates are learned)

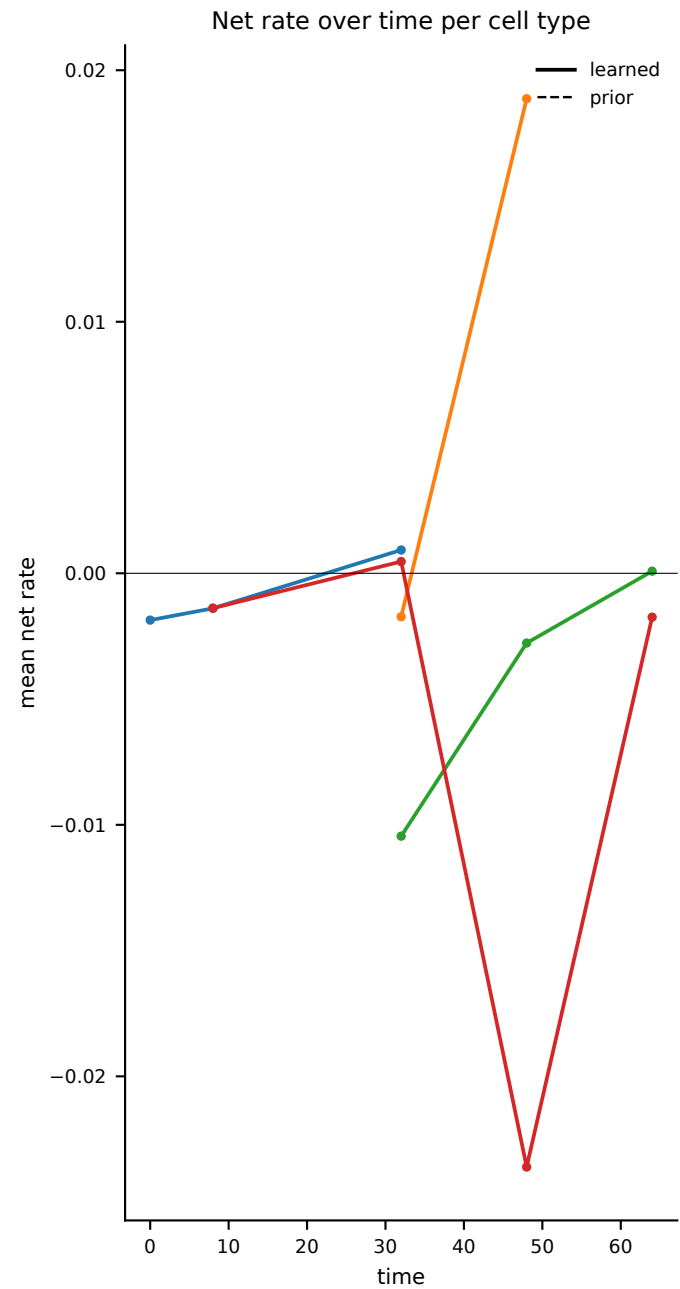
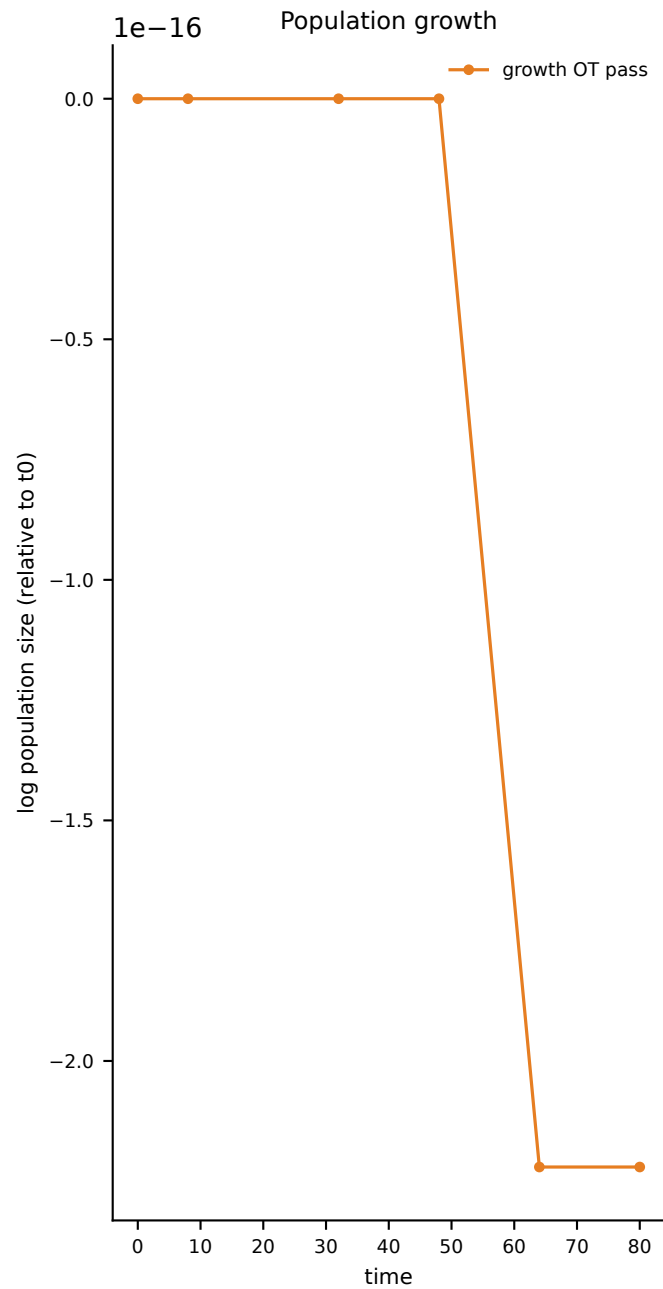


Mean net rate per cell type

Cell type	Anchor	Prior	Learned
precursors	—	—	-0.00141
intermediate	—	—	0.00699
chromaffin	—	—	-0.000738
sympathoblasts	—	—	-0.0104

4. Proliferation — population growth and drivers

Relative population size from the mean growth factor per interval; unless Data/population_sizes is given, its absolute level follows the prior.

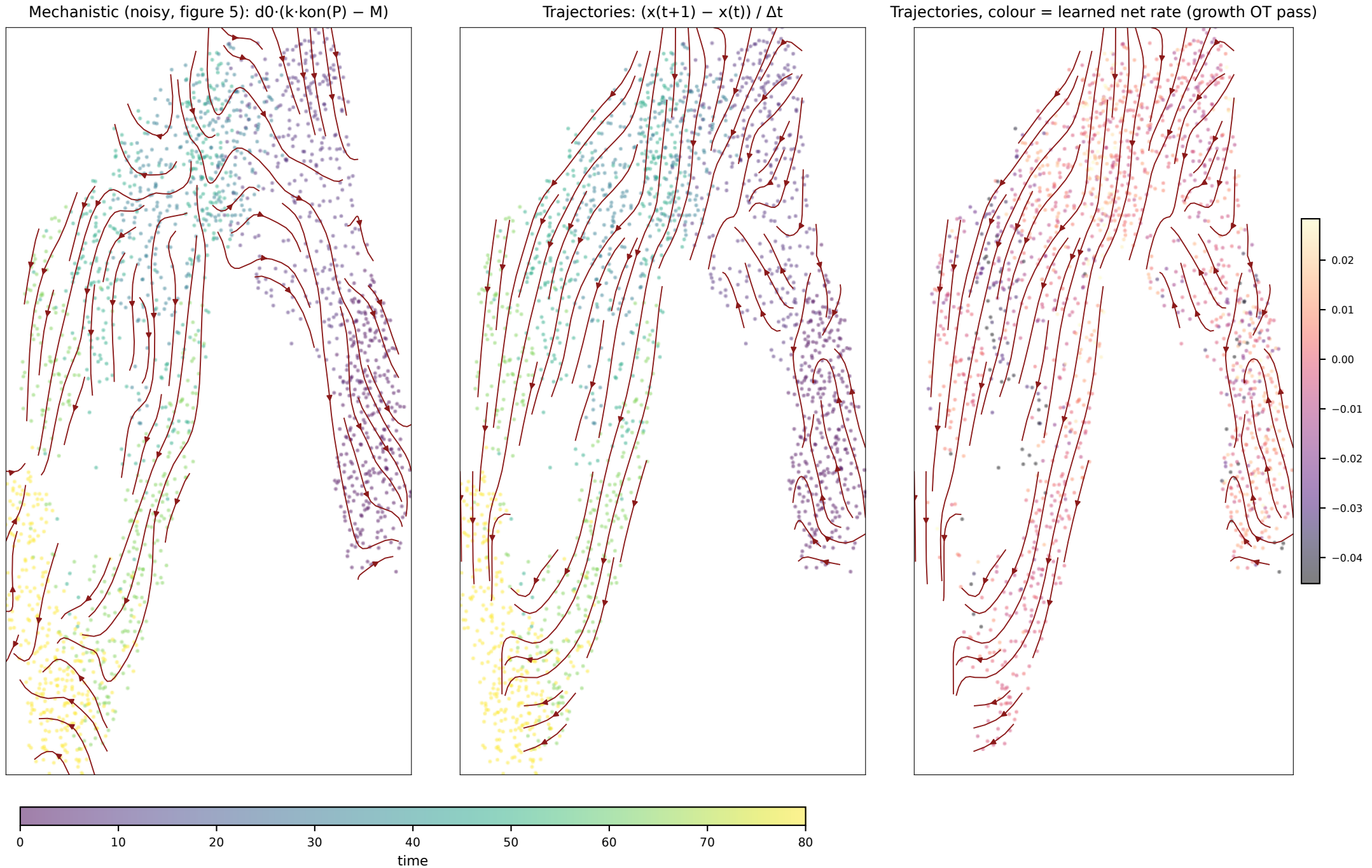


No proliferation MLP
(run infer_network_simul
--compute-proliferation)

precursors intermediate chromaffin sympathoblasts

5. Learned dynamics — mRNA velocity fields

1800 trajectory states; kNN-transition projection on a UMAP of log1p NB-sampled mRNA. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.30.



5. Learned dynamics — Proteins velocity fields

1800 trajectory states; kNN-transition projection on a UMAP of protein levels. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.54.

